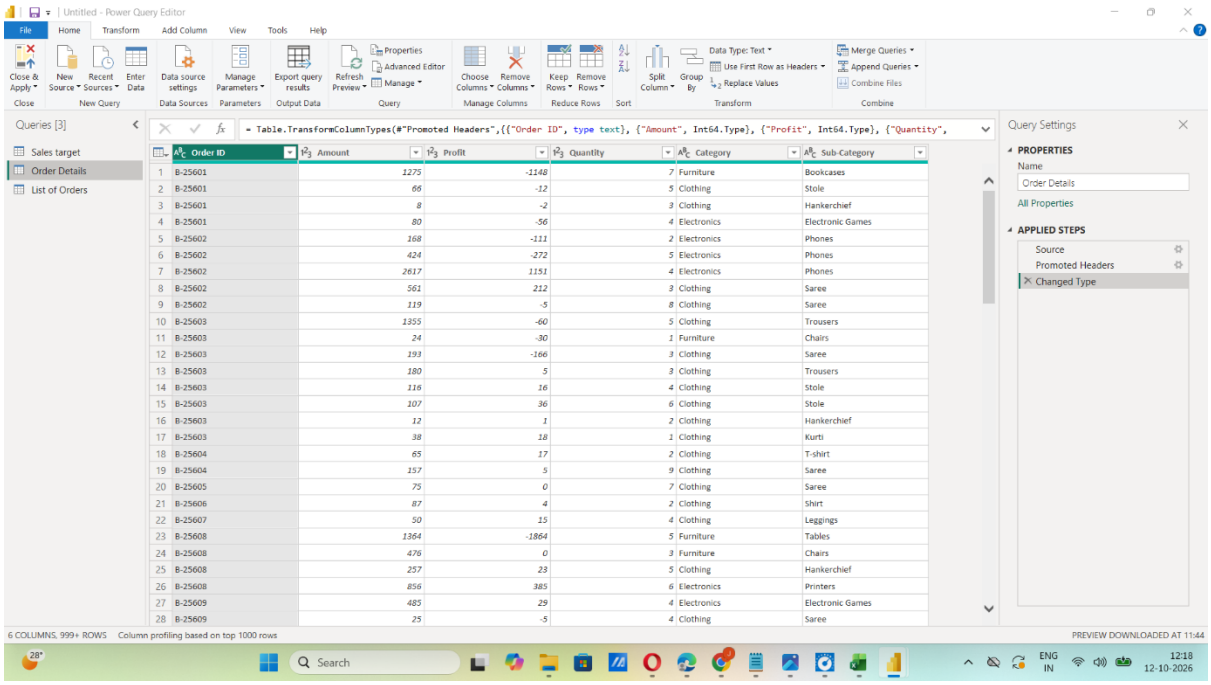
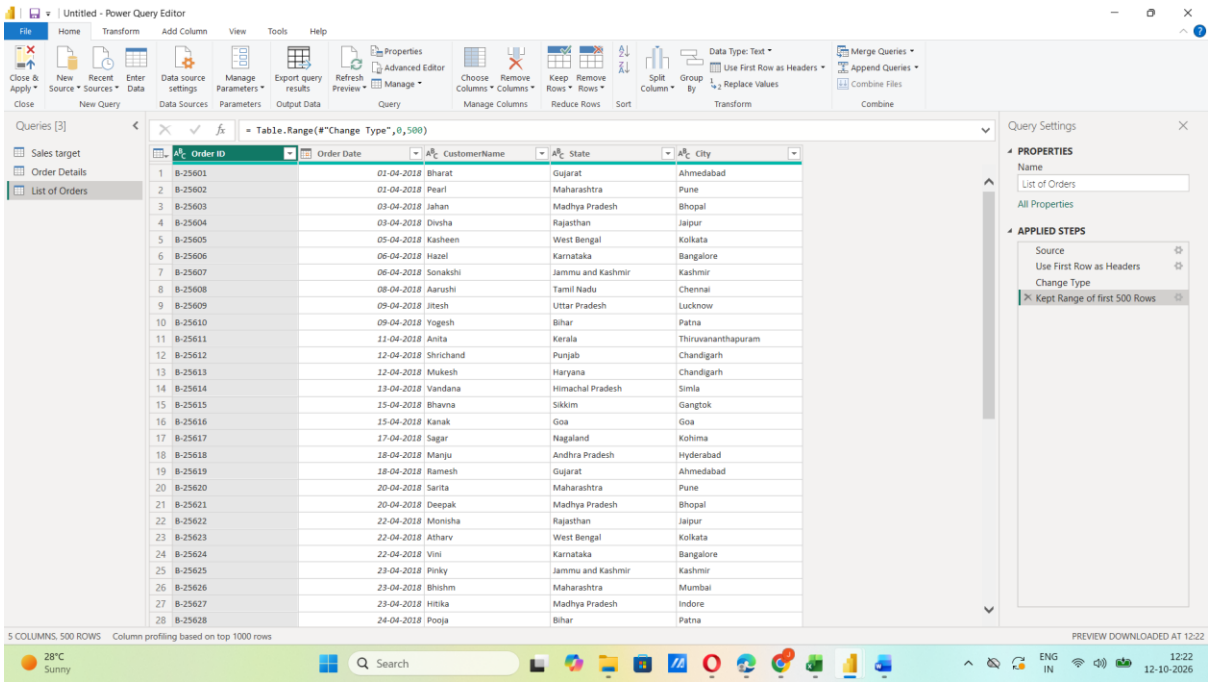
Observation

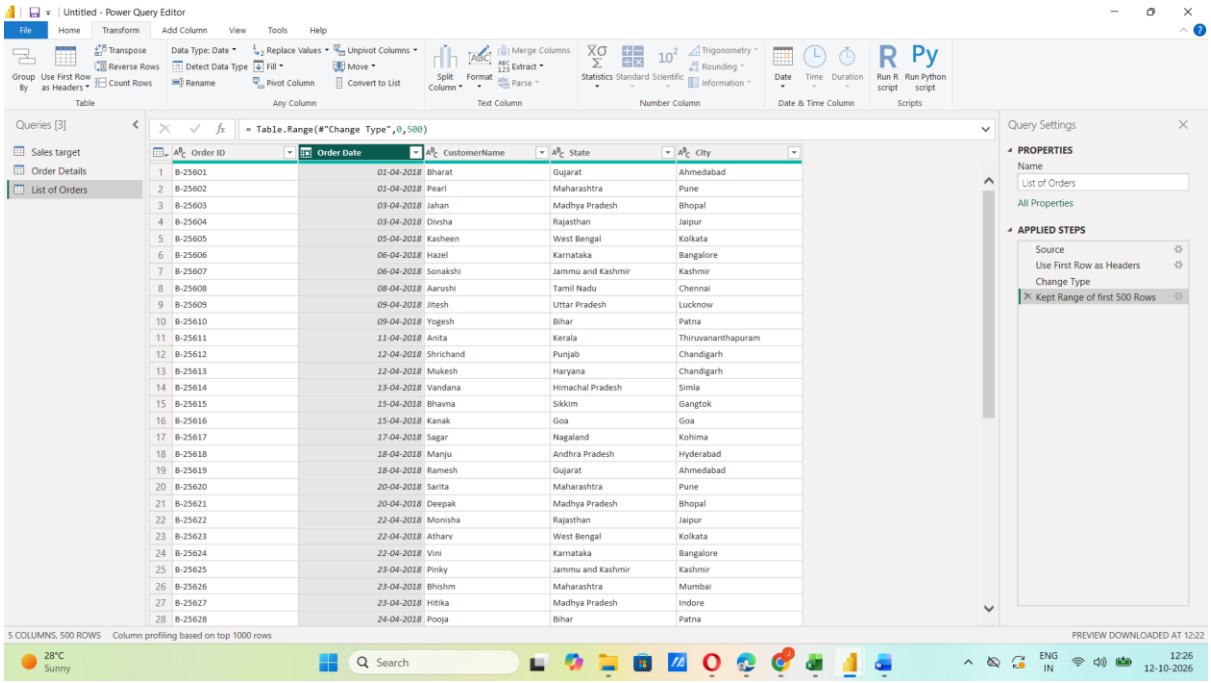
Step 1: Imported the three CSV files — **List of Orders.csv**, **Order Details.csv**, and **Sales target.csv** — into Power Query Editor for data transformation and analysis.



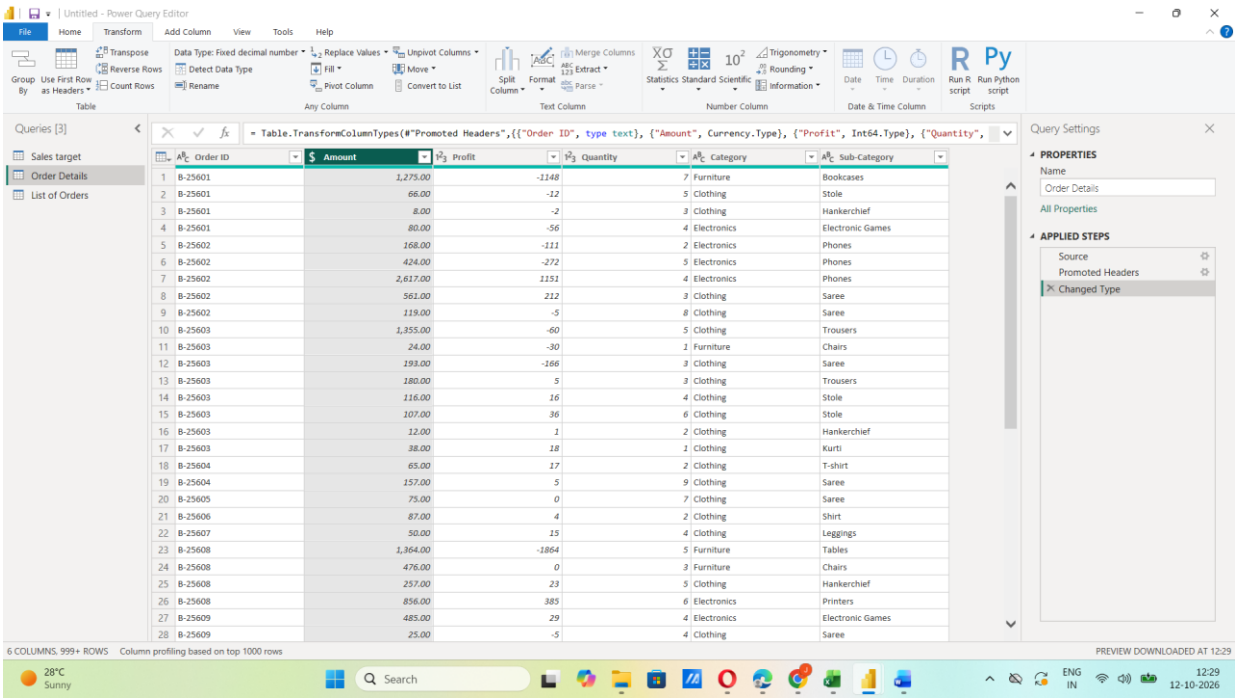
Step 2: Used the **Keep Top Rows** option to retain only the first 500 rows in the List of Orders table, as specified in the assignment..

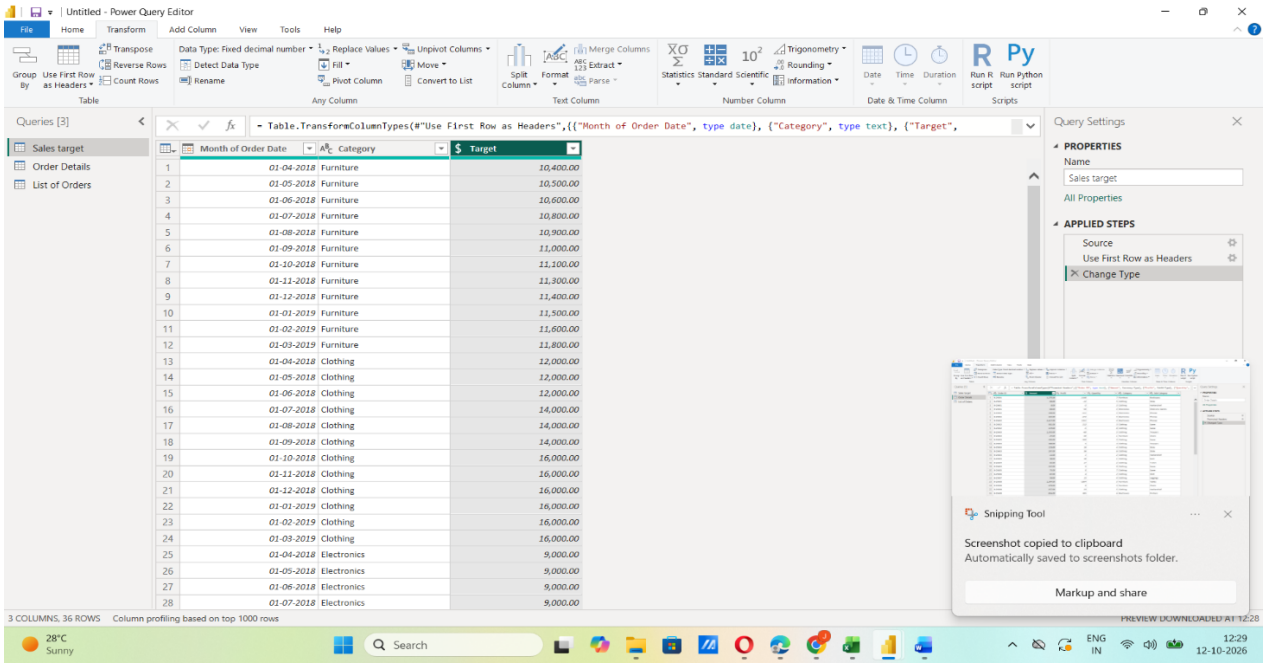


Step 3: Changed the data type of the Order Date column in the List of Orders table to **Date** to ensure that dates are recognized correctly.

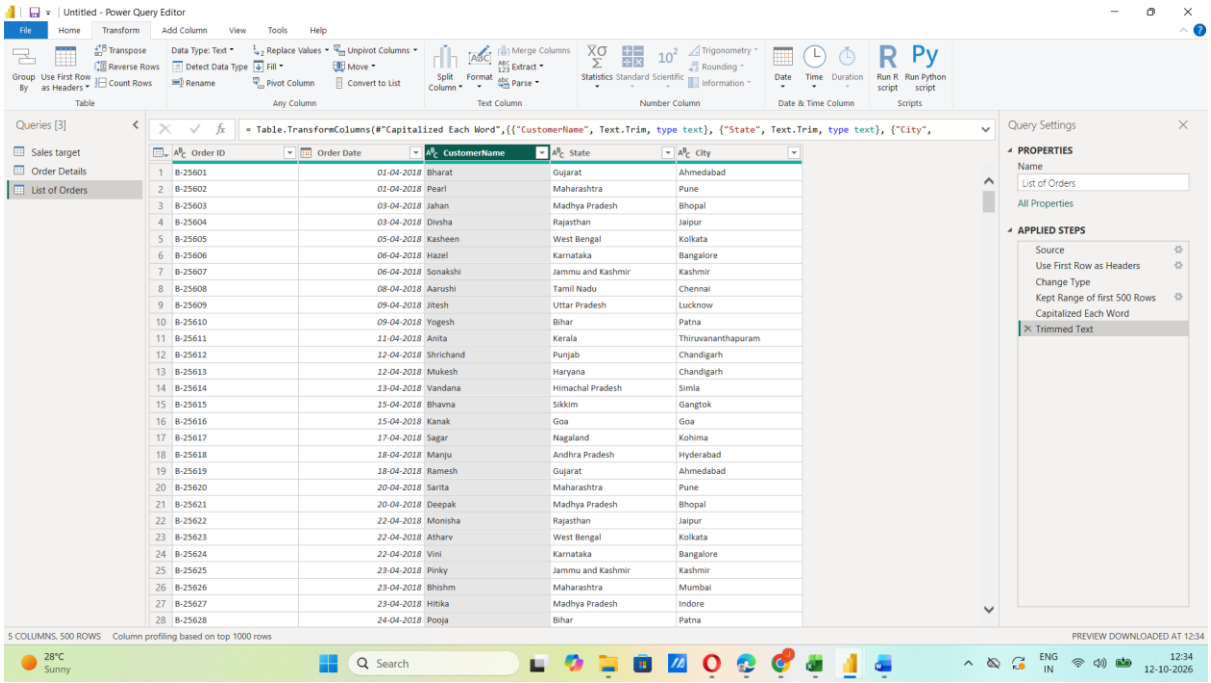


Step 4: Changed the data types of the Amount and Target columns to **Fixed Decimal Number** to ensure appropriate numeric formatting and calculations.

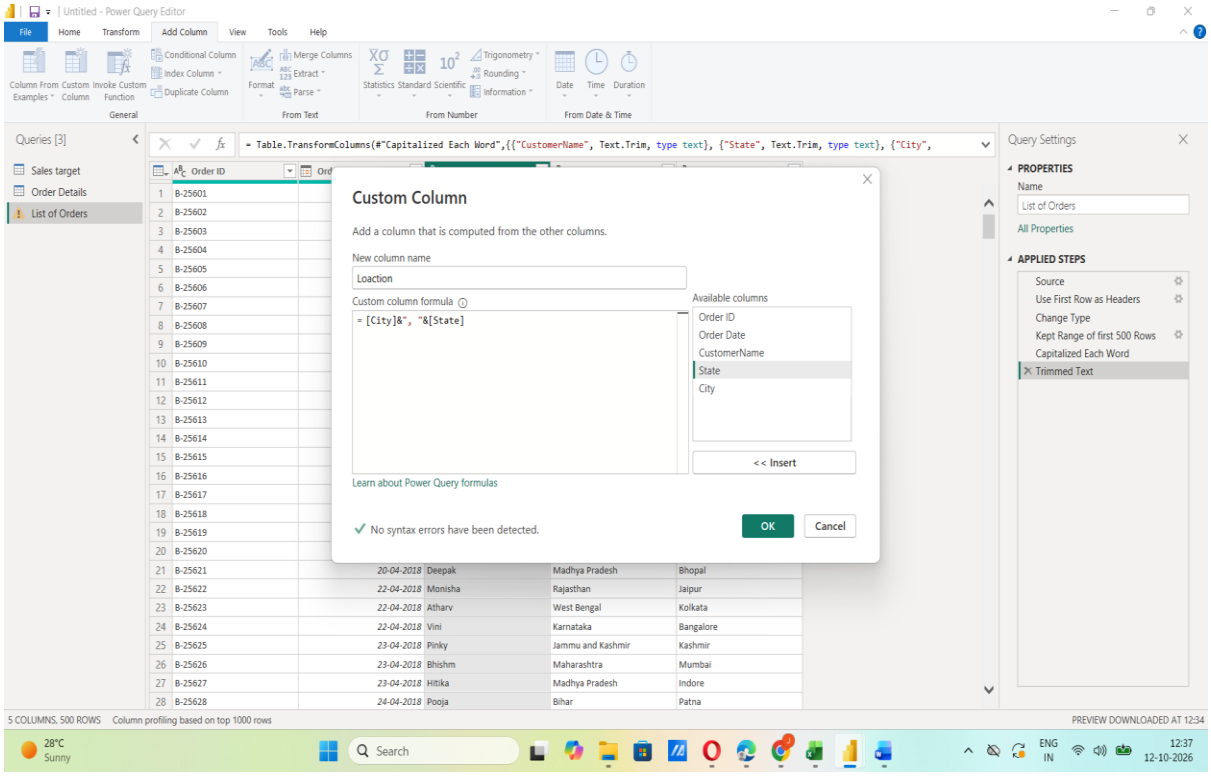


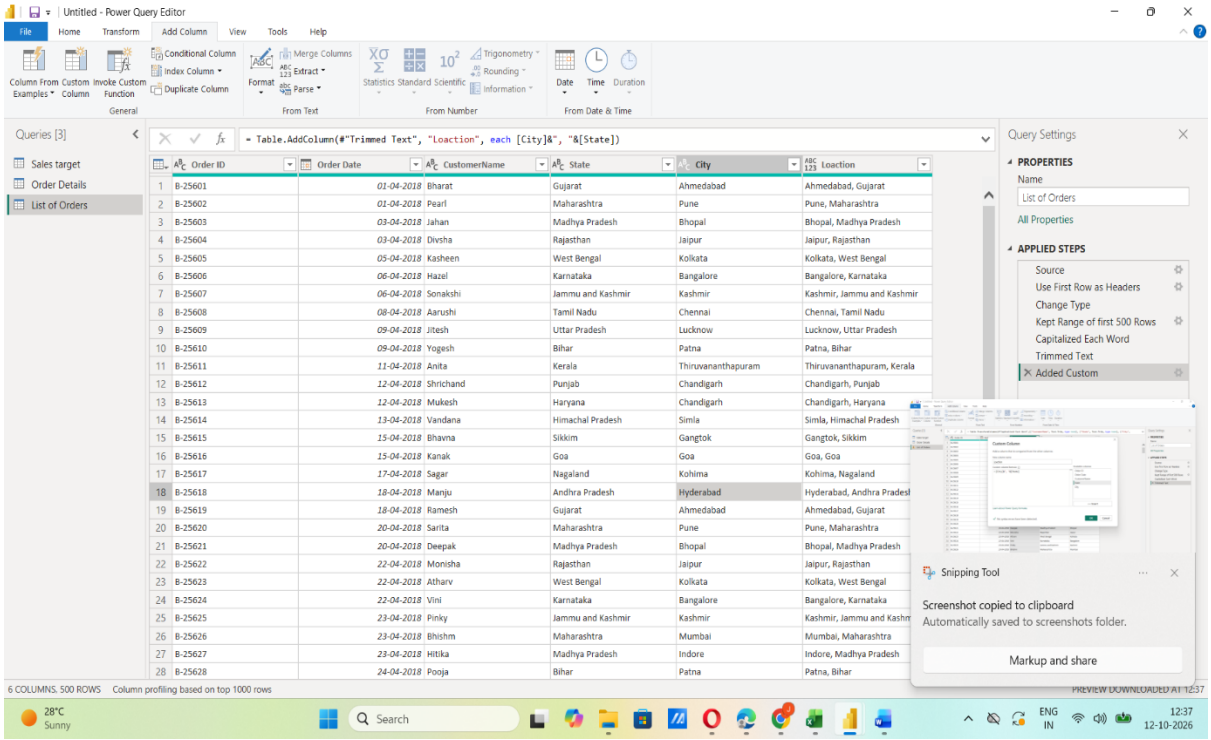


Step 5: Formatted the "CustomerName" column into proper case, ensuring consistent capitalization for each word.

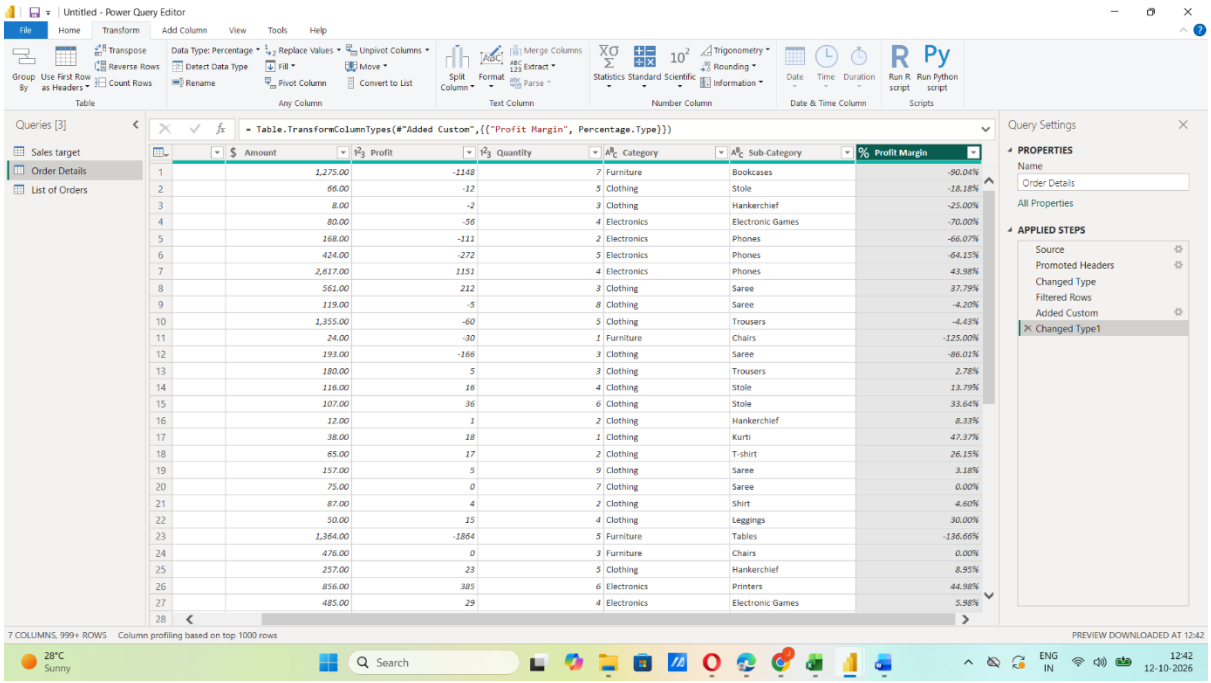


Step 6: Merged the "State" and "City" columns to create a new column named "Location" in the format ‘City, State’.



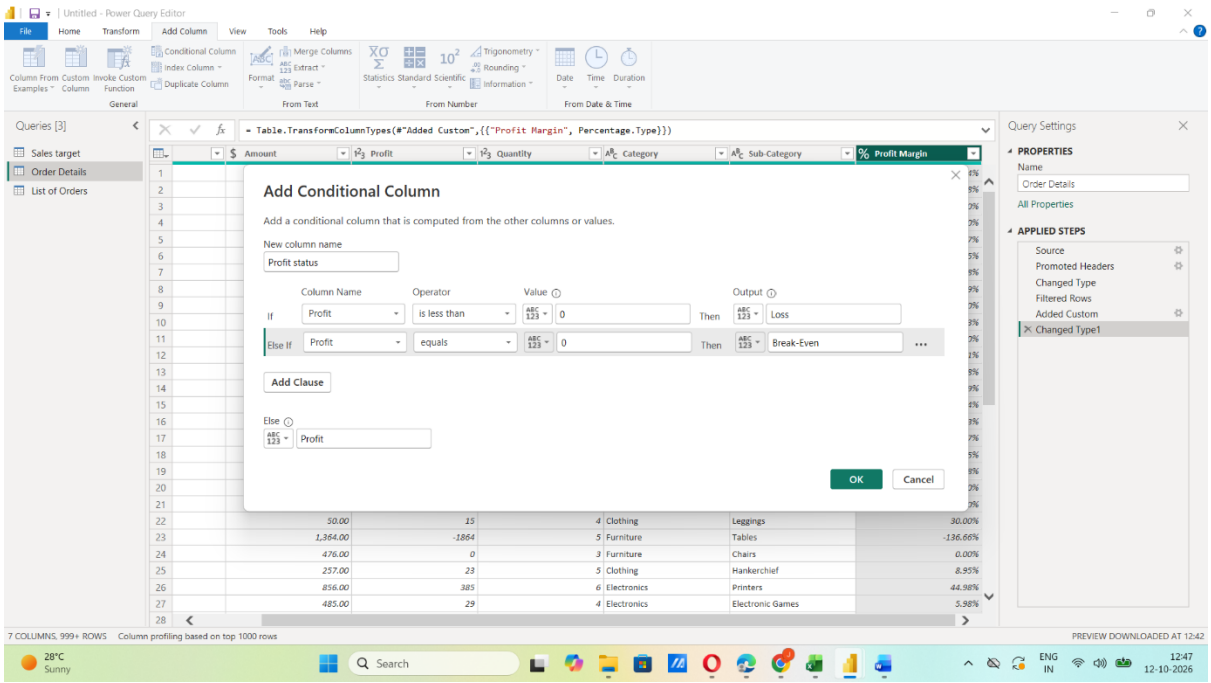


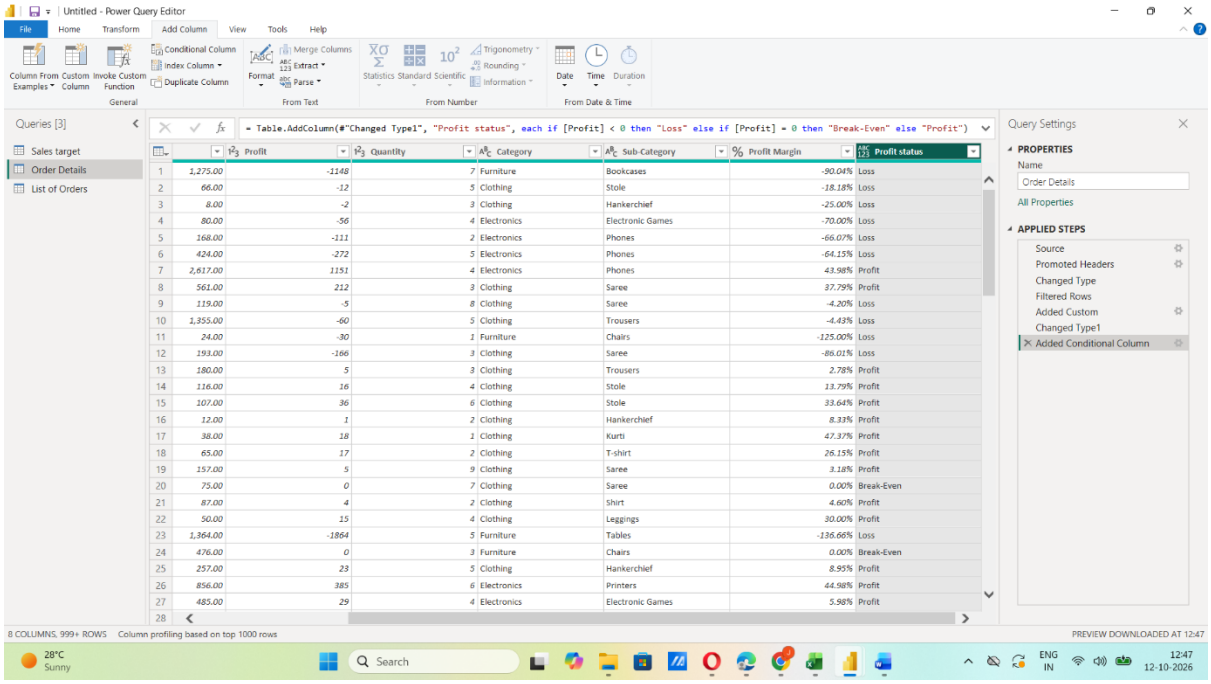
Step 7: Created a new custom column named "Profit Margin" as the percentage of "Profit" divided by "Amount". And changed data type into percentage



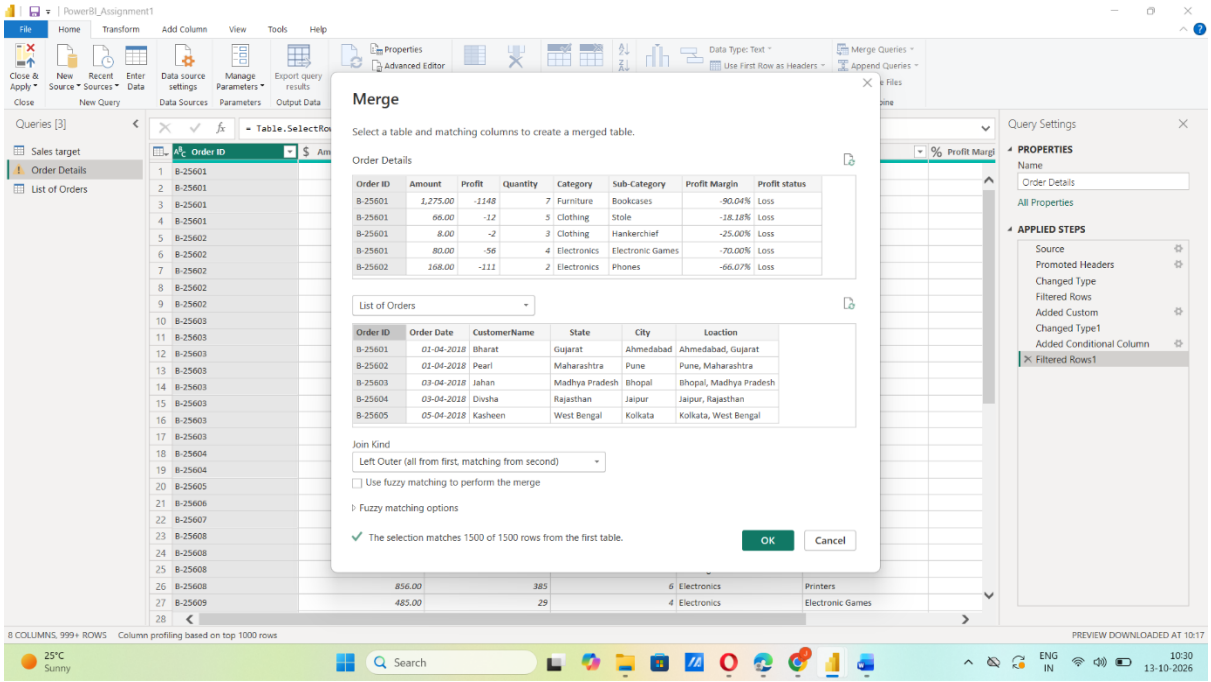
Step 8: Created a conditional column named Profit Status based on the Profit value:

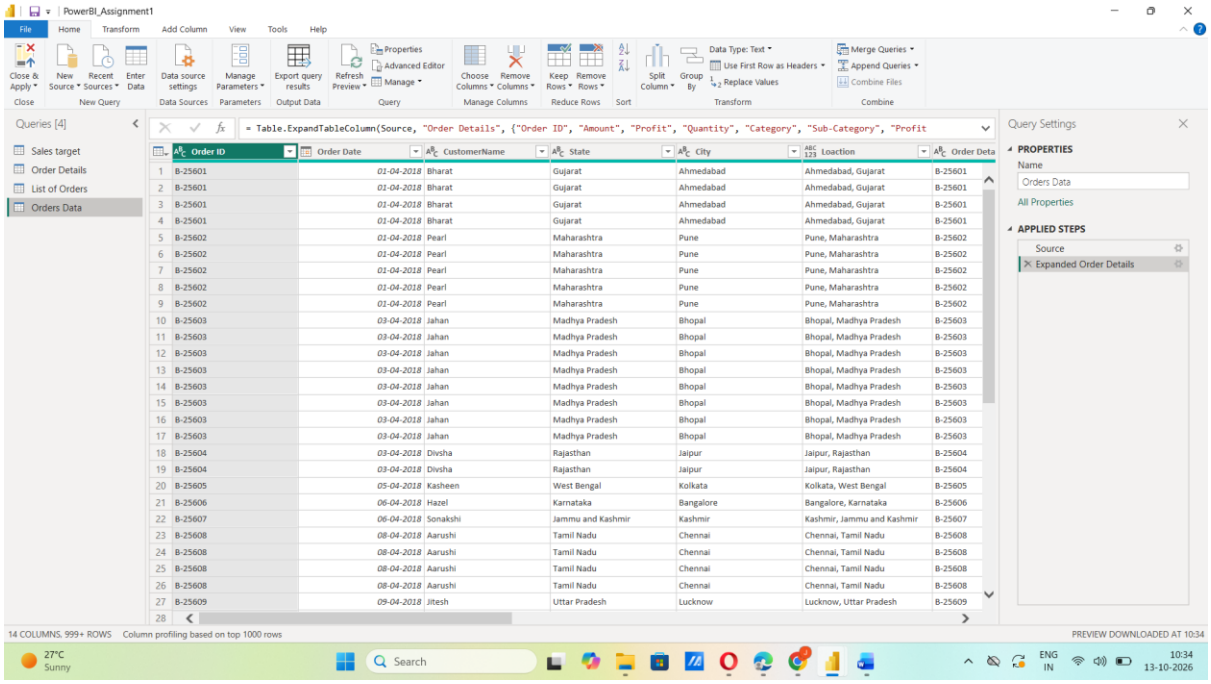
- If Profit is less than 0, the status is **Loss**.
- If Profit equals 0, the status is **Break-Even**.
- If Profit is greater than 0, the status is **Profit**.





Step 9: Used **Merge Queries as New** to combine the List of Orders and Order Details tables using Order ID as the matching column. Named the resulting table Orders Data.





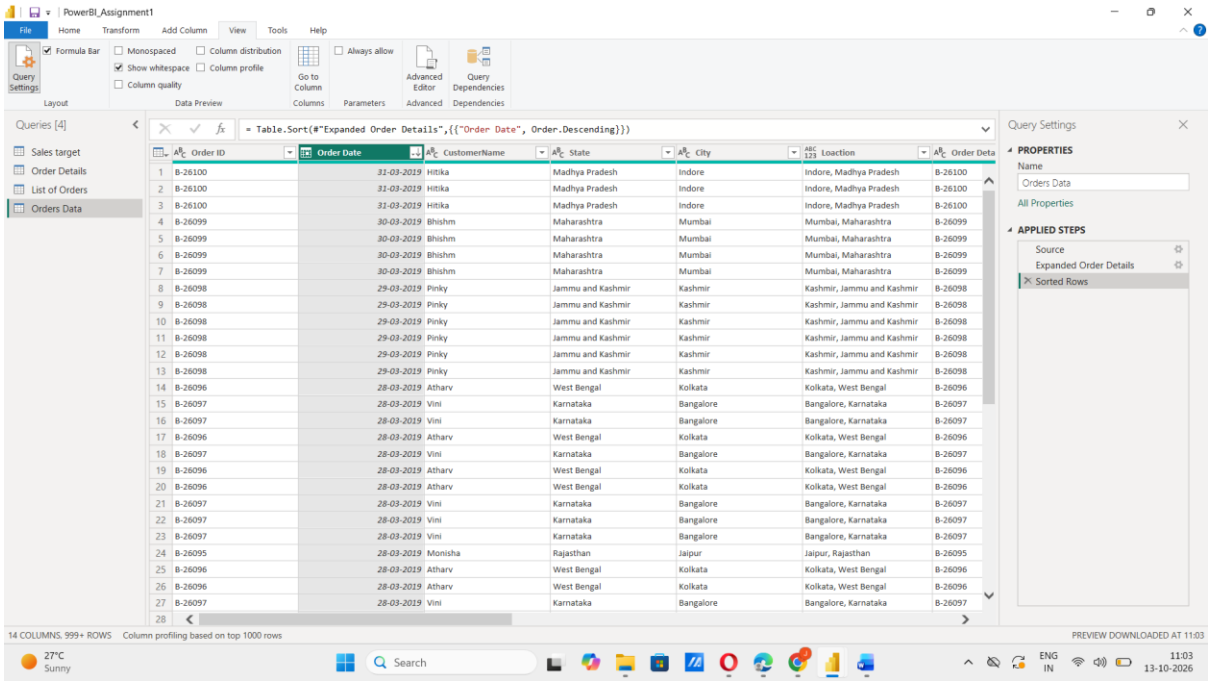
Step 10 : Handling Missing Data & Duplicate Data:

Used the Column Quality feature in Power Query to inspect missing values and errors. No missing values were identified in the data checked.

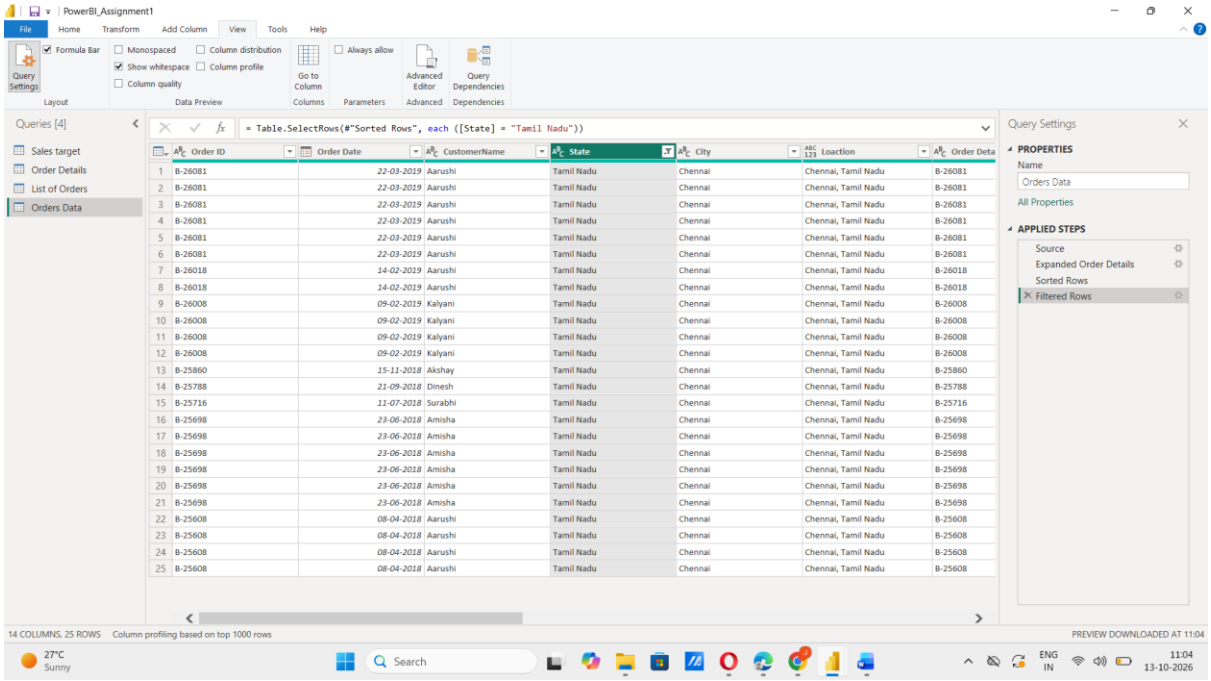
Checked for duplicate records using the Order ID, Category, and Amount columns. No duplicate combinations were identified in the data checked.

Step 11: Sorting and Filtering Data:

Sorted the orders by Order Date in descending order to analyze recent trends.



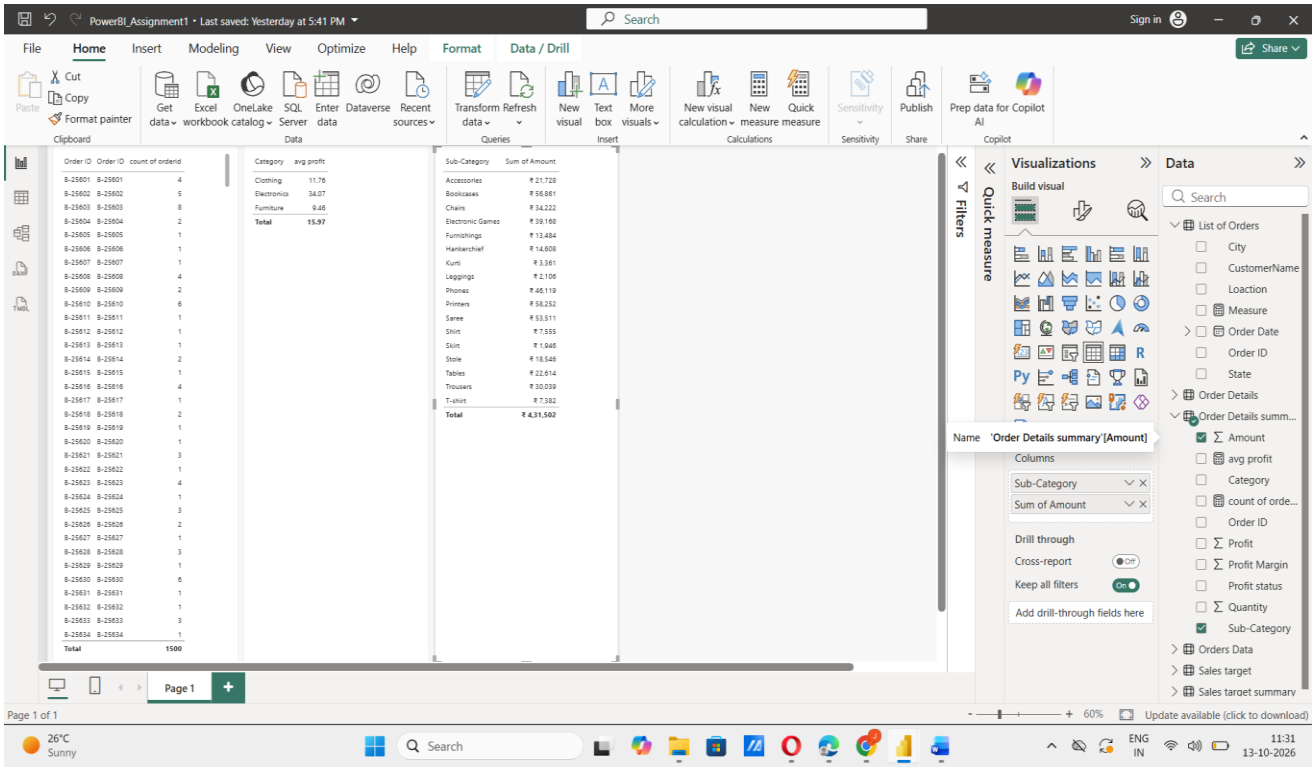
Filtered the orders to focus only on a specific state (e.g., Tamil Nadu) for regional analysis.



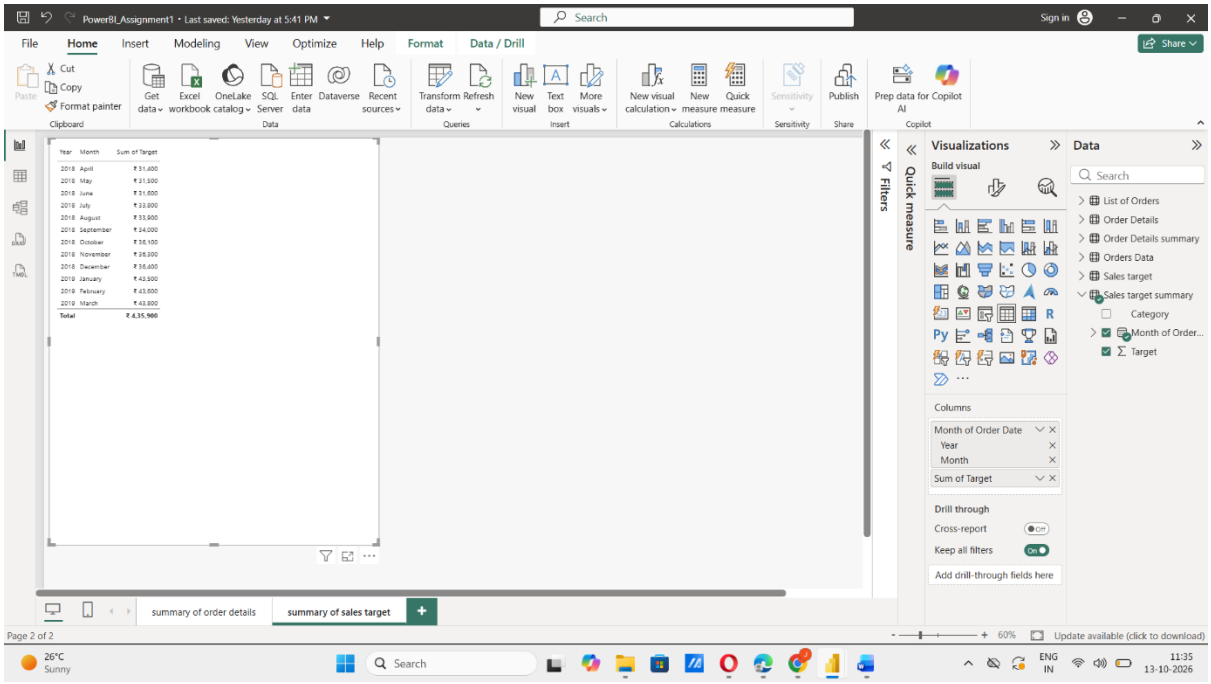
Step 11: Grouping and Aggregating Data:

Created a duplicate of order details table and named as order details summary to perform aggregate function. Created 3 table view.

1st table contains total count of order id, 2nd contains average profit by category and 3rd contains sum of amount by sub category.



Duplicated the “Sales Target” table ,named as sales target summary and created table view and I drag and dropped the order date and sum of target in table.



Step 12: Data modeling

Created a relationship between order details and list of orders by having order id in both column. Check relationships are active.

Same created relationship for sales target and order details by category . Check relationships are active.

